Neurobehavioral Factors in Eating Disorders

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Introduction

This report analyzes neurobehavioral factors associated with **self-perceived eating disorder status** among university students in Bangladesh. The dataset contains survey responses from **550 students** and includes measures related to cognitive restraint, emotional eating, food cue responsiveness, body image concerns, and related perceptions.

Primary outctome: `perception_eating_disorder` (binary; 0 = No, 1 = Yes).

Approach: I created composite domain scores and evaluated associations using group comparisons and logistic regression. A simplified model was selected for interpretability and comparable predictive performance.

Key findings (final model):

- **Affective (Emotional Eating)** was the strongest predictor (large odds ratio; substantial risk gradient across quartiles).
- **Cognitive Control/Restriction** showed a strong independent association with perceived eating disorder status.
- **Body Image Distress** (reverse-coded so **higher = worse** distress) remained a significant predictor.
- Predictive performance was good on a test set (**AUC ~ 0.82**), with balanced classification performance at a 0.5 probability threshold.

Practical implication: The model is suitable as a **screening-oriented decision-support tool** (not diagnostic), highlighting students who may benefit from follow-up assessment or targeted interventions (emotion regulation, healthy eating patterns, body image support).

Phase One: Ask

Eating disorders are shaped by a mix of brain and behavior factors, like how people handle their emotions, how they react to food cues, their efforts to restrict eating, and worries about body image. College students might be especially at risk because of the stress they face, big changes in their daily lives, and the pressure from social and environmental factors.

Business Task:

The goal of this analysis is to identify the **primary neurobehavioral drivers** of eating disorders among university students in Bangladesh. By analyzing **550** survey responses across seven critical factors, I aim to provide university health officials with data-driven insights to support the creation of targeted mental health interventions.

Key Stakeholders:

- **Primary Audience:** University administration, student health services, or mental health researchers.
- **Stakeholder Goal:** To design better prevention programs or early intervention strategies based on the "*brain-based models*" I mentioned in dataset description.

Success Metrics

1. **Dependent Variable (Target):** Self-perception about eating disorders.
2. **Independent Variables (Predictors):** Scores for Cognitive Control, Affective Value, Salience/Taste, Body Image, and Reward Processing.

Phase Two: Prepare

Data Source & Organization

- **Location:** The dataset is a flat file named `instagram_usage_lifestyle_1million.csv` obtained through Kaggle platform.
- **Structure:** It contains 1,000,000 rows (users) and 57 columns (attributes).
- **Organization:** Structured data with clear headers including Demographics, Lifestyle/Health, Daily Habits, and Instagram Usage.

  Note: All data is 100% synthetic — generated using statistical distributions and realistic correlations. No real user data was used or collected. Perfectly safe for research, education, prototyping, and Kaggle competitions.

ROCCC Assessment

To ensure the data is suitable for analysis, I performed a ROCCC assessment:

Criterion	Status	Assessment
Reliable	Medium	There are 550 responses. For a university-level study, this is a robust sample size, though i must acknowledge the bias inherent in self-reported surveys (students might under-report sensitive eating disorder symptoms).
Original	High	This is original data collected directly from students at Daffodil International University via email/social media.
Comprehensive	High	The dataset covers **seven critical factors** (Demographic, Cognitive Control, Affective Value, Salience, Body Image, Reward Processing, and Self-Perception). This suggests a wide set of features to test, which is excellent.
Current	High	The data was collected recently, from August to November 2024, making it highly relevant.
Cited	High	The dataset has it's source origin in it's description (Daffodil International University)

Privacy & Ethics

- **Context:** Since this deals with eating disorders (mental health), strict anonymity is vital.
- **Check:** I ensured the dataset does not contain **PII** (Personally Identifiable Information) like student IDs, names, or email addresses. If it does, my first step in the "Process" phase was to remove them.

Phase Three: Process

In this section I document *how the cleaned, analysis-ready dataset was produced*, the tools used, and the quality checks performed. The goal is reproducibility: a common reader should understand what was changed and why.

Tools Used

- **Excel / Google Sheets**: primary environment for cleaning, recoding, and aggregation.
- **R (later phases)**: analysis, modeling, validation, and reporting.

File management and versioning

- Raw dataset was kept unchanged.
- A separate cleaned file was created for analysis.
- Naming convention used:

  - `neurobehavioral_factors_in_eating_disorders_raw.csv` (original)
  - `neurobehavioral_factors_in_eating_disorders_clean.csv` (analysis-ready)

Analysis uses: `neurobehavioral_factors_in_eating_disorders_clean.csv`

Data cleaning steps performed in Excel

I used Excel because the dataset contains a resonable quantity of data.

Column standardization

- I verified all column headers were correctly labeled and consistent.
- Ensured no duplicate or blank column names.
- Removed trailing spaces and accidental duplicates.

Missing values and integrity checks

- Scanned each column for blank cells.
- Confirmed the dataset had one row per participant and no duplicated responses.
- Verified demographic fields had valid categories.

Recoding categorical survey responses into numeric scales

Survey response options (*Never, Seldom, Sometimes, Often*) were converted into numeric codes to allow statistical analysis.

Example frequency coding (as used in this case study):

- `Never` = 0
- `Seldom` = 1
- `Sometimes` = 2
- `Often` = 3

Recoding "days" response ranges into numeric codes

Items measured as frequency across a time window (*1–12 days, 13–27 days, Every day*) were also converted to numeric codes.
During analysis validation, I identified that these day-based variables were oriented such that **lower numeric values represented more frequent distress**. I corrected this later in the *analysis phase* by reverse-coding the composite body-image score.

Creating composite domain scores (aggregation)

To reduce dimensionality and support interpretable analysis, item-level variables were aggregated into composite domain scores. Aggregation was performed in **RStudio**, producing domain scores such as:

- `score_cognitive_control`
- `score_affective`
- `score_salience`
- `score_habit`

- `score_body_image`
- `score_risk_perception` (outcome variable)

Aggregation rule: row-wise mean (not sum)

Outcome variable preparation

The outcome column `Perception_EatingDisorder` was converted into a binary numeric variable:

- `0` = does not perceive having an eating disorder
- `1` = perceives having an eating disorder

Final processed dataset used in R

The final dataset imported into R includes:

- Demographic variables
- Numeric-encoded item responses
- Composite domain scores
- Binary outcome variable

File used for analysis: `neurobehavioral_factors_in_eating_disorders_clean.csv`

Reproducibility note

Because processing was done in Excel, formulas and mapping tables are not automatically tracked in code. To improve reproducibility, this project stores:

- the final analysis-ready dataset (`neurobehavioral_factors_in_eating_disorders_clean.csv`)
- exported result tables (`tables/`)
- saved model objects (`models/`)
- generated figures (`figures/`)

Analyze (Exploration, Statistical Testing, and Modeling)

This section answers the core question:

Which neurobehavioral domains are most strongly associated with self-perceived eating disorder status, and how well can a simple, interpretable model predict ED perception?

I follow an evidence-building sequence:

1) Verify outcome and distributions

2) Describe domain scores

3) Test group differences (bivariate)

4) Fit and simplify a logistic regression model (multivariable)

5) Evaluate performance on unseen data (test set)

6) Choose an operational threshold and build stakeholder-friendly risk summaries

Load libraries

```r
# Libraries for data work + plots + tables
library(tidyverse)
library(janitor)
library(knitr)
library(kableExtra)

# Libraries for modeling and evaluation
library(rsample)
library(broom)
library(pROC)
library(yardstick)

set.seed(123)  # makes results reproducible
```

Import the cleaned dataset

This case study starts analysis from a cleaned dataset (categorical responses already converted into numeric codes). I set a single source of truth for the cleaned dataset path.

```r
glimpse(df_clean)
```

```
## Rows: 550
## Columns: 57
## $ age_range                                        <chr> "18-24", "25-34", "1~
## $ gender                                           <chr> "Male", "Female", "M~
## $ division_of_origin                               <chr> "Rajshahi", "Chattog~
## $ current_education_level                          <chr> "Bachelor's degree 3~
## $ employment_status                                <chr> "Student", "Self-emp~
## $ marital_status                                   <chr> "In a relationship",~
## $ eat_less_on_weight_gain                          <dbl> 3, 3, 3, 3, 2, 3, 3,~
## $ eat_less_at_mealtime                             <dbl> 0, 3, 0, 3, 0, 3, 3,~
## $ refuse_food_weight_concern                       <dbl> 2, 1, 3, 0, 1, 3, 0,~
## $ monitor_food                                     <dbl> 2, 3, 1, 2, 3, 2, 1,~
## $ eat_slimming_foods                               <dbl> 0, 1, 2, 2, 0, 0, 2,~
## $ eat_less_after_overeating                        <dbl> 3, 3, 3, 3, 3, 2, 0,~
## $ eat_less_to_prevent_weight_gain                  <dbl> 1, 3, 3, 2, 3, 0, 3,~
## $ avoid_snacks_between_meals_for_weight_concern    <dbl> 3, 0, 2, 3, 3, 0, 1,~
## $ avoid_evening_eating_for_weight_concern          <dbl> 1, 2, 1, 2, 3, 0, 0,~
## $ consider_weight_when_eating                      <dbl> 3, 1, 0, 1, 2, 2, 3,~
## $ eat_when_unoccupied                              <dbl> 2, 2, 3, 0, 0, 3, 1,~
## $ eat_when_irritated                               <dbl> 2, 3, 3, 2, 2, 0, 0,~
## $ eat_when_depressed_or_discouraged                <dbl> 2, 3, 3, 0, 2, 1, 0,~
## $ eat_when_lonely                                  <dbl> 3, 1, 3, 0, 1, 0, 3,~
## $ eat_when_someone_let_you_down                    <dbl> 3, 3, 2, 2, 2, 2, 2,~
## $ eat_when_angry                                   <dbl> 0, 0, 3, 3, 1, 1, 3,~
## $ eat_when_expecting_bad                           <dbl> 1, 1, 3, 3, 0, 3, 0,~
## $ eat_when_anxious                                 <dbl> 1, 3, 2, 2, 1, 2, 2,~
## $ eat_when_things_go_wrong                         <dbl> 1, 3, 0, 0, 1, 1, 3,~
## $ eat_when_frightened                              <dbl> 2, 3, 2, 3, 1, 1, 2,~
```

```
## $ eat_when_disappointed                            <dbl> 1, 2, 3, 3, 2, 0, 2,~
## $ eat_when_emotionally_upset                       <dbl> 0, 3, 1, 1, 0, 0, 3,~
## $ eat_when_bored_or_restless                       <dbl> 0, 2, 0, 1, 1, 0, 3,~
## $ eat_more_if_food_tasty                           <dbl> 3, 3, 3, 3, 2, 3, 3,~
## $ eat_more_if_food_smells_or_looks_good            <dbl> 0, 2, 0, 2, 2, 3, 2,~
## $ eat_when_see_delicious_food                      <dbl> 2, 1, 3, 0, 1, 3, 0,~
## $ eat_delicious_food_immediately                   <dbl> 2, 3, 1, 2, 1, 2, 1,~
## $ desire_to_buy_from_bakery                        <dbl> 0, 1, 2, 2, 0, 0, 2,~
## $ desire_to_buy_from_snack_bar_or_cafe             <dbl> 3, 3, 3, 3, 3, 2, 0,~
## $ desire_to_eat_when_see_others_eating             <dbl> 1, 3, 3, 2, 3, 0, 3,~
## $ resist_delicious_food                            <dbl> 3, 0, 2, 2, 3, 0, 1,~
## $ eat_more_when_see_others_eating                  <dbl> 1, 2, 1, 2, 3, 0, 0,~
## $ eat_when_preparing_meal                          <dbl> 3, 1, 0, 1, 3, 2, 3,~
## $ eat_specific_foods_habitually                    <dbl> 3, 3, 0, 2, 3, 0, 0,~
## $ location_triggers_habitual_eating                <dbl> 3, 2, 2, 3, 3, 2, 2,~
## $ automatic_eating_when_experiencing_strong_emotion <dbl> 2, 0, 0, 3, 1, 2, 0,~
## $ realize_after_eating_out_of_habit                <dbl> 0, 0, 0, 3, 3, 1, 1,~
## $ days_desire_for_flat_stomach                     <dbl> 2, 1, 3, 1, 1, 2, 2,~
## $ days_felt_fat                                    <dbl> 3, 1, 1, 1, 2, 2, 1,~
## $ days_weight_affected_self_judgment               <dbl> 1, 1, 1, 1, 1, 0, 2,~
## $ days_shape_affected_self_judgment                <dbl> 3, 0, 2, 1, 1, 3, 0,~
## $ days_dissatisfied_with_weight                    <dbl> 1, 2, 1, 1, 1, 2, 1,~
## $ days_dissatisfied_with_shape                     <dbl> 1, 3, 1, 2, 1, 2, 0,~
## $ days_uncomfortable_to_see_own_body               <dbl> 1, 0, 3, 2, 3, 0, 2,~
## $ days_uncomfortable_others_seeing_your_shape      <dbl> 1, 0, 1, 0, 1, 2, 2,~
## $ days_tried_limit_food_to_control_shape_or_weight <dbl> 2, 1, 3, 3, 1, 2, 2,~
## $ days_fasted_to_control_shape_or_weight           <dbl> 3, 2, 1, 1, 2, 2, 0,~
## $ days_excluded_food_to_control_shape_or_weight    <dbl> 1, 2, 1, 3, 1, 0, 2,~
## $ days_followed_rules_to_control_shape_or_weight   <dbl> 3, 0, 2, 1, 1, 3, 0,~
## $ days_fear_losing_control_over_eating             <dbl> 1, 2, 1, 1, 1, 2, 3,~
## $ perception_eating_disorder                       <dbl> 1, 1, 1, 1, 1, 0, 1,~
```

Validate the outcome variable

Before analysis, I confirm the outcome is binary (0/1) and check how many students are in each group.

- `0` = does not perceive an eating disorder
- `1` = perceives an eating disorder

Table 2: Outcome distribution: Perception of Eating Disorder (0 = No, 1 = Yes)

ED perception	Count (n)	Proportion
0	254	0.462
1	296	0.538

I confirm composite domain scores exist (or compute them)

To make the analysis easier to understand, I created domain scores by averaging related questions. This reduces many survey items into a few meaningful scores.

```r
df_scored <- df_clean %>%
  rowwise() %>%
  mutate(
    # Cognitive control / restriction behaviors
    score_cognitive_control = mean(c_across(c(
      eat_less_on_weight_gain,
      eat_less_at_mealtime,
      refuse_food_weight_concern,
      monitor_food,
      eat_slimming_foods,
      eat_less_after_overeating,
      eat_less_to_prevent_weight_gain,
      avoid_snacks_between_meals_for_weight_concern,
      avoid_evening_eating_for_weight_concern,
      consider_weight_when_eating
    )), na.rm = TRUE),

    # Emotional eating (affective triggers)
    score_affective = mean(c_across(c(
      eat_when_unoccupied,
      eat_when_irritated,
      eat_when_depressed_or_discouraged,
      eat_when_lonely,
      eat_when_someone_let_you_down,
      eat_when_angry,
      eat_when_expecting_bad,
      eat_when_anxious,
      eat_when_things_go_wrong,
      eat_when_frightened,
      eat_when_disappointed,
      eat_when_emotionally_upset,
      eat_when_bored_or_restless
    )), na.rm = TRUE),

    # Salience / food cue responsiveness
    score_salience = mean(c_across(c(
      eat_more_if_food_tasty,
      eat_more_if_food_smells_or_looks_good,
      eat_when_see_delicious_food,
      eat_delicious_food_immediately,
      desire_to_buy_from_bakery,
      desire_to_buy_from_snack_bar_or_cafe,
      desire_to_eat_when_see_others_eating,
      resist_delicious_food,
      eat_more_when_see_others_eating,
      eat_when_preparing_meal
    )), na.rm = TRUE),

    # Habit / automatic eating
    score_habit = mean(c_across(c(
      eat_specific_foods_habitually,
      location_triggers_habitual_eating,
      automatic_eating_when_experiencing_strong_emotion,
```

```r
      realize_after_eating_out_of_habit
    )), na.rm = TRUE),

    # Body image distress (days-based items)
    score_body_image = mean(c_across(c(
      days_desire_for_flat_stomach,
      days_felt_fat,
      days_weight_affected_self_judgment,
      days_shape_affected_self_judgment,
      days_dissatisfied_with_weight,
      days_dissatisfied_with_shape,
      days_uncomfortable_to_see_own_body,
      days_uncomfortable_others_seeing_your_shape,
      days_tried_limit_food_to_control_shape_or_weight,
      days_fasted_to_control_shape_or_weight,
      days_excluded_food_to_control_shape_or_weight,
      days_followed_rules_to_control_shape_or_weight,
      days_fear_losing_control_over_eating
    )), na.rm = TRUE)
  ) %>%
  ungroup()
```

Reverse the body image score (higher = worse)

In this dataset, the "days" questions were coded so that **lower numbers** meant **more distress**. To interpret everything consistently (higher = worse), I reverse the composite body image score.

```r
# Find the maximum value used in the days coding (usually 3)
days_max <- max(df_scored$days_felt_fat, na.rm = TRUE)

# Reverse score so higher means more distress
df_scored <- df_scored %>%
  mutate(score_body_image_rev = days_max - score_body_image)
```

I do a quick check:

```r
df_scored %>%
  group_by(perception_eating_disorder) %>%
  summarise(mean_body_image_rev = mean(score_body_image_rev, na.rm = TRUE))
```

```
## # A tibble: 2 x 2
##   perception_eating_disorder mean_body_image_rev
##                        <dbl>               <dbl>
## 1                          0                1.47
## 2                          1                1.63
```

Summary statistics for the scores

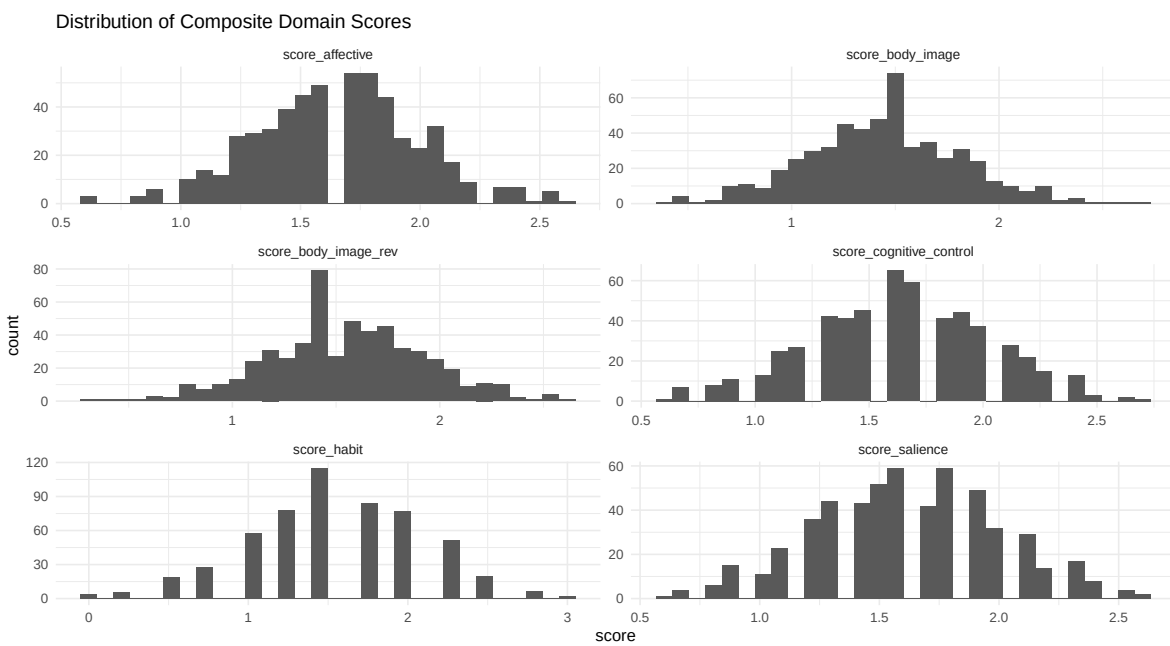
This gives a simple overview of average levels and variation.

```r
df_scored %>%
  select(starts_with("score_")) %>%
  summary()
```

```
##  score_cognitive_control score_affective  score_salience  score_habit
##  Min.   :0.600           Min.   :0.6154   Min.   :0.60   Min.   :0.000
##  1st Qu.:1.400           1st Qu.:1.4615   1st Qu.:1.30   1st Qu.:1.250
##  Median :1.600           Median :1.6923   Median :1.60   Median :1.500
##  Mean   :1.633           Mean   :1.6534   Mean   :1.62   Mean   :1.551
##  3rd Qu.:1.900           3rd Qu.:1.8462   3rd Qu.:1.90   3rd Qu.:2.000
##  Max.   :2.700           Max.   :2.6154   Max.   :2.60   Max.   :3.000
##  score_body_image score_body_image_rev
##  Min.   :0.3846   Min.   :0.3077
##  1st Qu.:1.1538   1st Qu.:1.3077
##  Median :1.3846   Median :1.6154
##  Mean   :1.4434   Mean   :1.5566
##  3rd Qu.:1.6923   3rd Qu.:1.8462
##  Max.   :2.6923   Max.   :2.6154
```

Plot the score distributions

These plots show whether scores look reasonably spread out (not all the same).



Compare ED=0 vs ED=1 (simple group test)

I check if average scores differ between groups.

Table 3: Group differences (t-tests): ED=0 vs ED=1

domain	mean_no	mean_yes	p_value
score_cognitive_control	1.468898	1.773986	0.0000000
score_affective	1.514537	1.772609	0.0000000
score_salience	1.485433	1.735135	0.0000000
score_habit	1.520669	1.577703	0.2203215
score_body_image_rev	1.465778	1.634615	0.0000005

I build the final logistic regression model

A logistic regression predicts a binary outcome (0/1). I use the three strongest domains we found earlier:

- Emotional eating (`score_affective`)
- Cognitive control (`score_cognitive_control`)
- Body image distress (`score_body_image_rev`)

Train/test split

```
split_obj <- initial_split(df_scored, prop = 0.7, strata = perception_eating_disorder)
train <- training(split_obj)
test  <- testing(split_obj)
```

Fit model (train set)

```
m_simple <- glm(
  perception_eating_disorder ~ score_affective + score_cognitive_control + score_body_image_rev,
  data = train,
  family = binomial()
)

summary(m_simple)
```

```
##
## Call:
## glm(formula = perception_eating_disorder ~ score_affective +
##     score_cognitive_control + score_body_image_rev, family = binomial(),
##     data = train)
##
## Coefficients:
##                         Estimate Std. Error z value Pr(>|z|)
## (Intercept)              -9.8092     1.1288  -8.690  < 2e-16 ***
## score_affective           2.6377     0.4150   6.355 2.08e-10 ***
## score_cognitive_control   2.2754     0.3518   6.468 9.96e-11 ***
## score_body_image_rev      1.2327     0.3228   3.819 0.000134 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 529.99  on 383  degrees of freedom
## Residual deviance: 404.06  on 380  degrees of freedom
## AIC: 412.06
```

```
##
## Number of Fisher Scoring iterations: 4
```

Odds ratios table (easier interpretation)

Table 4: Final model odds ratios (higher OR = higher risk)

term	estimate	conf.low	conf.high	p.value
score_affective	13.980811	6.376340	32.565786	0.000000
score_cognitive_control	9.732103	4.990067	19.892653	0.000000
score_body_image_rev	3.430477	1.846086	6.564169	0.000134

Evaluate the model (test set)

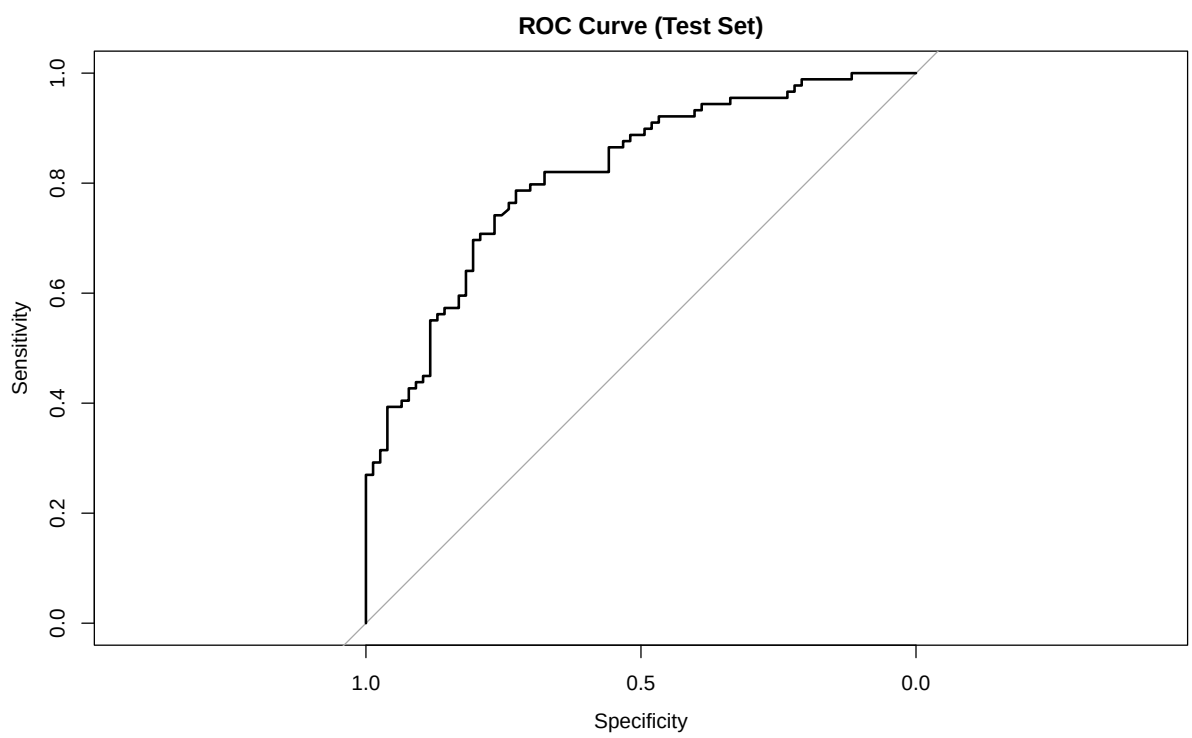
Predict probabilities

```r
test <- test %>%
  mutate(pred_prob = predict(m_simple, newdata = test, type = "response"))
```

ROC curve and AUC

```r
roc_obj <- roc(test$perception_eating_disorder, test$pred_prob)
auc(roc_obj)
```

```
## Area under the curve: 0.8156
```



Confusion matrix (threshold = 0.50)

```r
final_t <- 0.50

truth <- factor(test$perception_eating_disorder, levels = c(0, 1))
pred  <- factor(if_else(test$pred_prob >= final_t, 1L, 0L), levels = c(0, 1))

table(Truth = truth, Predicted = pred)
```

```
##      Predicted
## Truth  0  1
##     0 56 21
##     1 21 68
```

Key metrics (event = ED=1)

Table 5: Model performance on test set (threshold = 0.50)

threshold	accuracy	sensitivity	specificity	precision
0.5	0.746988	0.7640449	0.7272727	0.7640449

Risk summary (quartiles)

This makes the model easier to explain: it shows how predicted risk changes from low scores to high scores.

Table 6: Predicted risk: low (Q1) vs high (Q4) for each key domain

aff_q1	aff_q4	cog_q1	cog_q4	body_q1	body_q4
0.2840481	0.763228	0.2604202	0.7653002	0.3632788	0.6630594

Final takeaways

Key findings from the final model

- **Emotional eating** (`score_affective`) is the strongest predictor: students with higher emotional-eating scores are much more likely to report that they believe they have an eating disorder.
- **Cognitive control / restriction** (`score_cognitive_control`) is also strongly related to ED perception: more dieting/restriction behaviors are linked to higher likelihood of ED perception.
- **Body image distress** (`score_body_image_rev`) adds independent value: students with higher body image distress are more likely to report ED perception.

Model quality (how well it predicts)

- **AUC (ROC curve)** shows how well the model separates ED=1 from ED=0 across all thresholds.
- At a **0.50 threshold**, we get a balanced trade-off between:
  - **Sensitivity** (catching ED=1 cases)
  - **Specificity** (correctly ruling out ED=0 cases)

Important limitation (interpretation rule)

This is a **cross-sectional survey**:

- The results show **association**, not causation.
- The model is best used as a **screening support tool**, not a diagnostic instrument.

Share (Visualizations and Key Findings)

In this phase I communicate the results clearly to different stakeholders (university health services, administration, and researchers).
The goal is to answer: **What did I find, why does it matter, and what should be done next?**

Key findings (headline statements)

- Students with higher **emotional eating** scores are substantially more likely to report ED perception.
- **Cognitive restriction/control** and **body image distress** are also independently associated with ED perception.
- The final logistic regression model shows **good discrimination** on the test set (AUC reported below).
- At a 0.50 cutoff, the model balances sensitivity and specificity reasonably well for general reporting.

This report includes five "stakeholder-ready" artifacts:

- Outcome distribution (context)
- Group differences plot (ED=0 vs ED=1)
- Final model odds ratios (interpretation)
- Model performance (ROC/AUC + confusion matrix metrics)
- Risk gradient plot (what changes from low to high scores)

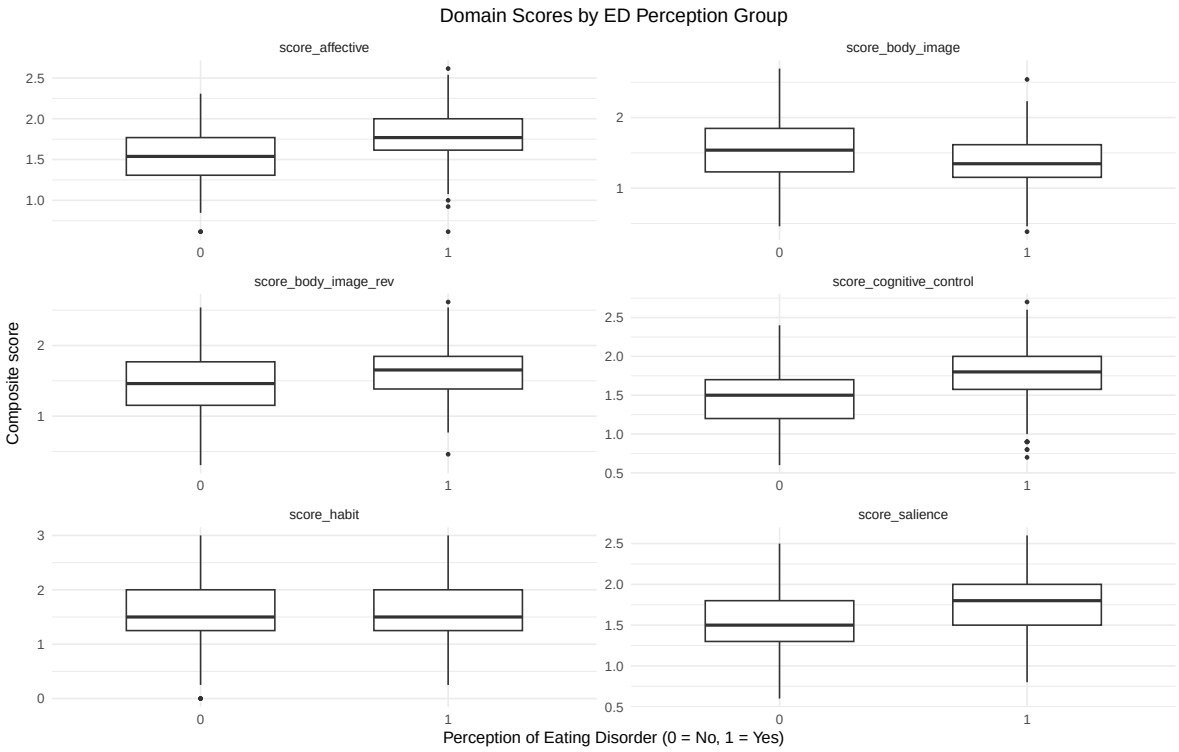
Visual 1 - Outcome distribution (context)

Table 7: Outcome distribution (0 = No, 1 = Yes)

ED (0/1)	Count (n)	Percent (%)
0	254	46.2
1	296	53.8

Visual 2 — Group differences in domain scores (ED=0 vs ED=1)

This figure shows whether domain scores differ between groups before modeling.



Visual 3 — Final model results (Odds Ratios)

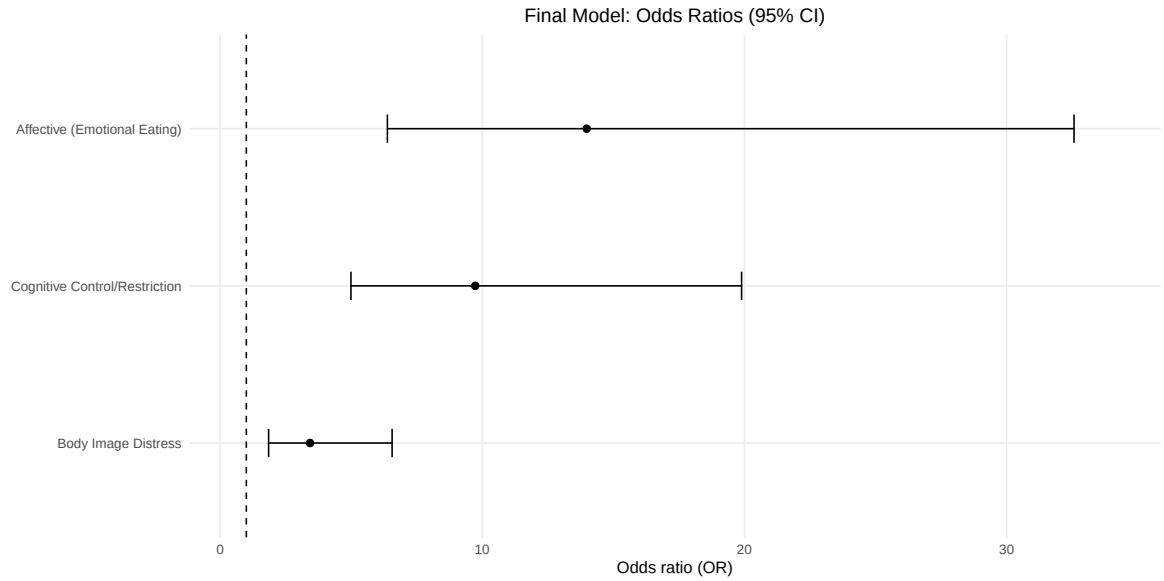
Odds ratios communicate "how much higher" the odds are when a score increases.

Odds ratio table

Table 8: Final model odds ratios (95% CI)

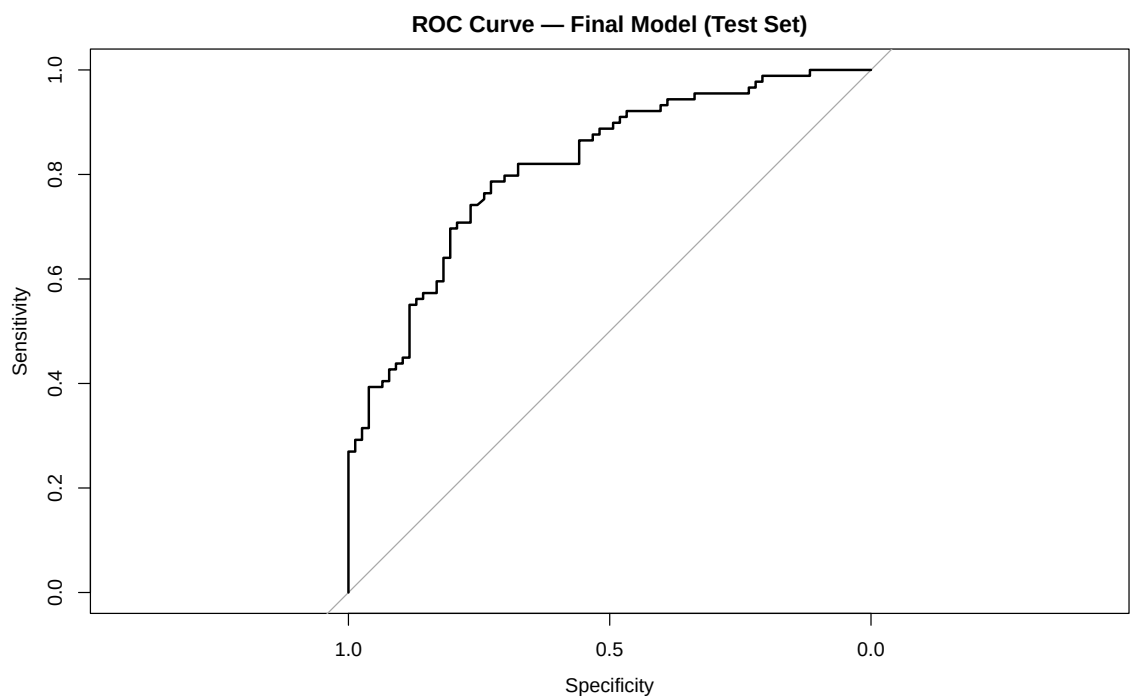
Predictor	OR	CI low	CI high	p-value
Affective (Emotional Eating)	13.98	6.38	32.57	<0.001
Cognitive Control/Restriction	9.73	4.99	19.89	<0.001
Body Image Distress	3.43	1.85	6.56	<0.001

Odds ratio plot



Visual 4 — Model performance (ROC/AUC + metrics)

ROC curve and AUC



Confusion matrix and core metrics (threshold = 0.50)

```
stopifnot(exists("test"))
stopifnot("pred_prob" %in% names(test))
```

```r
final_t <- 0.50
truth <- factor(test$perception_eating_disorder, levels = c(0, 1))
pred  <- factor(if_else(test$pred_prob >= final_t, 1L, 0L), levels = c(0, 1))

cm <- table(Truth = truth, Predicted = pred)
cm
```

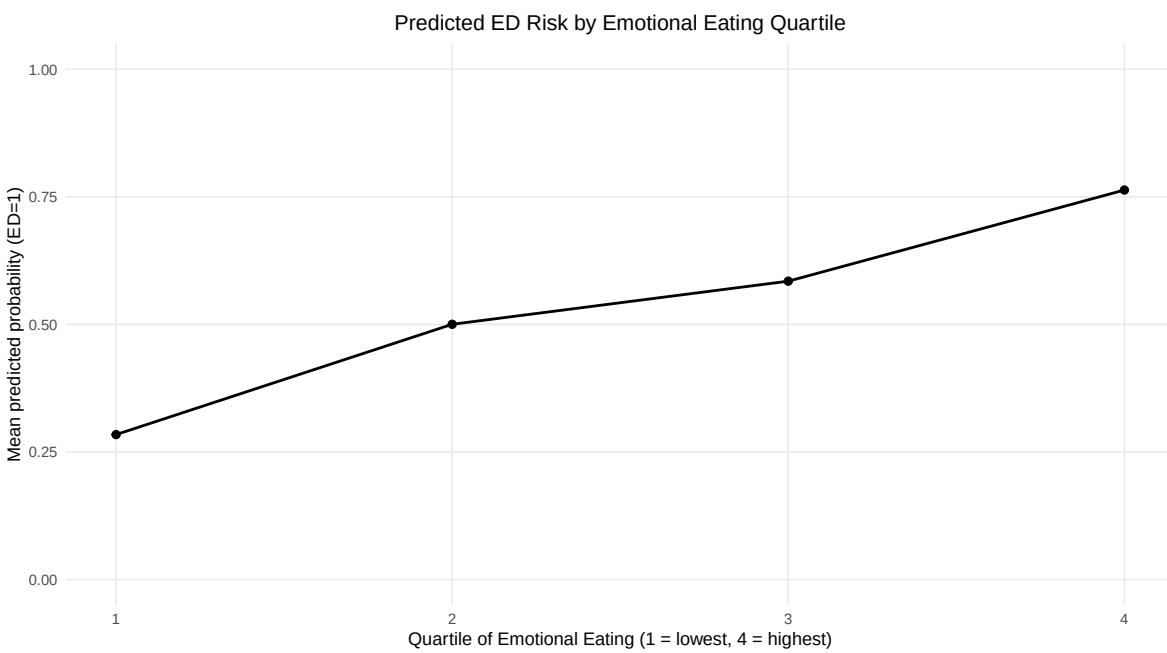
```
##      Predicted
## Truth  0  1
##     0 56 21
##     1 21 68
```

Table 9: Test set performance (threshold = 0.50, event = ED=1)

Threshold	Accuracy	Sensitivity	Specificity	Precision
0.5	0.747	0.764	0.727	0.764

Visual 5 — Risk gradient (stakeholder interpretability)

This plot shows how predicted risk changes from low to high scores (quartiles).



Act (Conclusions, Recommendations, and Next Steps)

- **What is the final conclusion?**
- **How can stakeholders use these insights?**
- **What should be done next, and why?**

Final conclusion (what the analysis supports)

Based on the domain-level modeling and validation:

- **Emotional eating (Affective domain)** is the strongest predictor of self-perceived eating disorder status.
- **Cognitive control/restriction** is also strongly associated with ED perception.
- **Body image distress** adds meaningful independent information.
- The final model shows **good discrimination** on the test set (AUC reported in Phase Five), indicating it can separate higher-risk vs lower-risk students better than chance.

Interpretation rule: These are **associations** from a cross-sectional survey. They do not prove that one factor causes another.

What these findings can support

- **Early screening** (identify students who may benefit from follow-up support)
- **Program design** (target the most influential domains)
- **Resource allocation** (prioritize interventions with the highest expected impact)

This model should be treated as **decision support**, not a clinical diagnostic tool.

Recommended actions (specific and practical)

Recommendation 1 — Target emotional eating (highest-impact domain)

- **What to do:** Implement workshops or counseling modules focused on emotion regulation and stress coping skills (e.g., coping strategies for anxiety, low mood, anger, disappointment).
- **Why:** Emotional eating had the strongest association and steepest risk gradient in the model.
- **Who should implement:** University counseling services in partnership with student affairs.
- **When:** Pilot within one academic term; evaluate uptake and engagement.

Recommendation 2 — Address restrictive eating behaviors and dieting cycles

- **What to do:** Provide psychoeducation about restrictive dieting patterns and encourage balanced eating strategies (aligned with nutrition guidance).
- **Why:** Cognitive control/restriction remained an independent predictor in the final model.
- **Who should implement:** Health services and nutrition educators.
- **When:** Integrate into student wellness programs, orientation sessions, and counseling referrals.

Recommendation 3 — Create body image support programming

- **What to do:** Offer body image resilience sessions and support groups, focusing on self-judgment and dissatisfaction frequency.
- **Why:** Body image distress added predictive value even after including other domains.
- **Who should implement:** Counseling services and trained facilitators.
- **When:** Offer recurring sessions; prioritize high-stress periods (exams).

Risks and limitations (what could go wrong)

- **False positives:** some students flagged by the model may not have an eating disorder.
- **False negatives:** some students may not be flagged even if at risk.
- **Self-report bias:** students may under-report sensitive behaviors.
- **Generalizability:** data is from one university, so performance may change elsewhere.

Next steps (recommended improvements)

External validation

Run the same model on a second student sample (different university or year cohort).
Goal: confirm the model generalizes.

Longitudinal data collection

Collect follow-up responses over time.
Goal: test whether these domains predict future outcomes (stronger evidence than cross-sectional).

Improve measurement reliability

If any domain had low internal consistency, refine item selection or apply psychometric validation.
Goal: ensure composites measure the intended constructs.

Build a simple screening rubric

Translate predicted probabilities into "Low / Medium / High" risk bands.
Goal: make the results easier to apply in practice.

Act deliverables (what is produced at the end)

- A validated, interpretable logistic regression model (`m_simple`)
- A stakeholder-friendly summary of key predictors and performance
- A recommended threshold policy for balanced vs screening use
- A set of actionable intervention recommendations tied directly to model findings